A-A-51371
31 March 1986

COMMERCIAL ITEM DESCRIPTION

RESUSCITATOR AND ASPIRATOR

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers a portable rescue breathing set and shall include a hand-operated resuscitator, a foot-operated aspirator and a carrying case. Shall be for use by medical personnel during emergency resuscitation of humans.

Salient characteristics.

Material and description. The resuscitator shall be designed for hand-operation and shall have the capability to administer either of both air and enriched oxygen concentrations, to re-establish and assist normal breathing during respiratory emergencies occurring in a wide range of climatic conditions.

The resuscitator assembly shall include a self-filling compressible butyl rubber or silicon bag. The bag shall have a minimum volume capacity of 1.6 liters, and shall be designed to control bag recovery, and shall be designed/shaped for single hand support/pump operation.

The proximal end of the bag shall be fitted with a non-rebreathing patient valve that is contained in a suitable plastic housing having a standard 15/22 mm connector at the patient end which is in accordance with ANSI Z79.8. The distal end of the bag shall be fitted with an adapter ring that shall be connectable to the manufacturer's optional accumulator. The adapter ring shall include a check valve inlet and a nipple for connection of a low pressure oxygen tube. The valve shall be capable of admitting either ambient air or oxygen enriched air. The nipple shall not provide a direct path into the bag. It shall have a side outlet for oxygen to mix with air inside the adapter ring in proximity of the inlet valve. The complete adapter ring shall be capable of enriching the oxygen concentration of the air just prior to admission into the bag.

The resuscitator shall also include a reusable medium adult size face mask that shall consist of a suitable face seal, a suitably transparent front section which allows visual monitoring of patient breath content, and a connector. The connector shall be compatible with the bag connector and standard adapters specified in ANSI Z79.2 and Z79.6.

The foot-operated aspirator assembly shall have the capacity to remove secretions from the patient's airway to enable successful resuscitation efforts. The aspirator shall effectively function with an overflowing collection jar. The assembly shall be supported on a base that allows either foot or hand operation.

AMSC N/A FSC 6515

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The aspirator assembly shall consist of a foot-operated bellows assembly, a collection jar assembly, a 56-inch (142 cm) minimum length medical grade patient suction tube with a catheter connector, and a corrosion-resisting base to support the aspirator assembly.

The aspirator bellows assembly shall consist of a bellows pump with an internal spring hardened AISI 302 stainless steel spring for powering suction strokes. The top plate and bottom base shall be aluminum, hard coated in accordance with ASTM C84A. The bottom base shall contain a suitable corrosion resistant flapper disc valve and disc valve retainer, a suitable nipple connected to a component medical grade tube that shall have the other end connected to the jar assembly.

The jar assembly shall be made of inert plastic material and shall include a rubber-compound cover that shall have corrosion resistant connectors for the attachment of the bellows tube and the patient suction tube. The jar assembly shall be protected by a corrosion-resisting upside down "U" shaped protector. The jar shall have a minimum capacity of 12 ounces and be readily detachable/attachable for emptying of debris during emergencies.

The resuscitator and aspirator shall be housed in a component, non-reflective, dull colored, plastic carrying case with a durable, yet comfortable handle. The case shall be suitable for storage and transport. It shall have a minimum wall thickness of 1/8 inch (0.3 cm), and a structural rib and section divider design that provides for strength and ease of packing/unpacking. The case shall include provisions for securing the components in the case. It shall be designed to allow operation of the aspirator assembly without removal, and provide for quick removal of all components.

The complete assembly, packed in the carrying case, shall not weigh more than 11.2 pounds (5.1 kg), and shall have a maximum cube of 1.1 cu. ft. (29 liters).

Cleaning and sterilization. The resuscitator face mask, bag, valve(s), adapter ring, and aspirator jar, jar cover, cover connectors, bellows, bellows spring, bellows base, bellows tube, patient suction tube, and catheter connector shall be easy to disassemble, easy to clean after use, easy to sterilize using either ethylene oxide gas, or cold chemical disinfectant liquid agent(s), and designed to prevent misassembly. The aspirator assembly shall also be capable of pump flush cleaning without removing it from the carrying case.

Durability. All stainless steel parts shall be in accordance with the appropriate ANSI Standards. The following components shall meet accelerated aging requirements specified in Method 7221 of Federal Test Method Standard No. 601 face mask, resuscitator bag, suction jar cover, suction jar, and bellows

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Operation. The rescue breathing set shall perform as described below with no significant degradation after operating in climatic conditions from 0°F to 120°F (-18°C to 50°C) and relative humidity between 5 and 95 percent:

a. The non-breathing patient valve in the resuscitator shall prevent rebreathing of exhalant gas. It shall not have a pressure relief device.

b. The resuscitator bag shall have a minimum stroke displacement volume of 900 cc. It shall fully self recover at a rate which allows 30 cycles per minute of operation.

c. The face mask seal shall maintain softness, provide effective fit, and minimize leakage.

d. The resuscitator shall operate with an oxygen flow rate as high as 15 LPM without lockup or jamming of non-rebreathing valve in the inhalation position.

e. With the resuscitator receiving an average tidal volume of 1000 cc, an oxygen flow rate of 15 LPM, and a cycle rate of 12 cycles per minute, the resulting low supplementary oxygen concentration shall be a minimum of 35 percent without an optional accumulator, and the resulting high oxygen concentration shall be a minimum of 85 percent with the manufacturer's optional accumulator.

f. The aspirator bellows assembly shall be capable of creating a partial vacuum, a minimum of 8 inches (20 cm) of Hg during successive strokes of the bellows, and shall also be capable of developing a suction when the jar is overflowing with secretions. If overflowing, the overflow shall be sucked from the top of the jar into the bellows and exhausted through the flapper disc valve in the bellows base.

Instructions The carrying case shall contain manual(s) describing the theory of operation, operating instructions, assembly/disassembly instructions, a list of replacement parts by manufacturer's catalog number, descriptive sketches as appropriate, statement of warranty, and manufacturer's and distributor's names, addresses, and phone numbers.

Workmanship. The rescue breathing set shall be free from defects which detract from its appearance or impair its serviceability.

Inspection. The contractor shall perform adequate inspection to assure compliance with this Description. The government reserves the right to perform any of the inspections set forth herein. A lot shall be any group of units from which the Government selects samples for inspection. The contractor shall be responsible for maintaining a quality assurance program and shall be responsible to replace/repair any and all defective units, or lots of units that equal or exceed 2.5 percent defective units, that are rejected by the Government. Visual examinations that do not require the use of special laboratory appliance/procedure shall not be restricted to defects listed herein. Test determination by technical means of physical and chemical properties of the item characteristics shall be as listed under "Tests:".

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Visual examination.

1. Component missing or not as specified.
2. Component not free of including (but not restricted to) cracks, checking, breaks, and deformed areas of the collection containers, and/or tubing.
3. Unit not clean
4. Collection container not readily removable.
5. Blistering, checking, peeling, scratching, or bare areas of the finish.
6. Any screw not drawn up tight, rivet not properly installed.
7. Service data not included.
8. Identification marking missing, incomplete, illegible, or not properly attached.
9. Aspirator not operable while in carrying case when case is opened, and patient end of suction tube is withdrawn from case, and no other action taken by the user/inspector.
10. Metal parts not free of corrosion.

Tests:

Drainage bottle. The drainage bottle shall be subjected to an internal vacuum of 12 inches ± 20 percent of Hg for a period of one (1) minute without failure.

Drop test. The unit, with all components inserted in place within the case, shall be dropped from a height of 18 inches onto dry earth. The case shall not open and components shall not be damaged or dislodged from their proper location within the case.

Water resistance. The unit, in its case (case closed in upright position, shall be subjected to a shower at the rate of 3 inches per hour for a period of 5 minutes. The contents shall be examined to determine if moisture has entered the case.

Humidity. At a relative humidity of 95 ± 5 percent, cycle the unit as follows: Raise temperature linearly from 80°F. to 115°F. in 6 hours, maintain at 120°F. for 6 hours, decrease temperature linearly from 120°F. to 80°F. in 6 hours, maintain at 80°F. for 6 hours. Then test the unit for performance as specified hereunder after a 1/2 minute operating time

Performance. The unit (in its case, shall be placed on a level surface and operated in accordance with the manufacturer's instructions (which shall be included in the carrying case. The following determinations shall be made after operating at least 1/2 minute

Vacuum at bellows end of patient tube shall be as specified.

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Insert patient end of the tube in water and continue to pump until after water is overflowing. There shall still be a negative pressure at the patient end of the tube while pumping and water is overflowing.

Following the manufacturers' instructions, empty the full collection container and have back in operation, allowing a maximum of 15 seconds from last stroke before emptying to first stroke after emptying.

Resuscitator Performance. Each test herein shall be performed twice, once in ambient temperature for 5°F. (±5°) and humidity of 10 percent (±5 percent) after standing in 5°F. temperature for 3 hours, and once in ambient temperature of 115°F. (±5°) and humidity of 90 percent (±5 percent) after standing in 115°F. temperature for 3 hours.

A single one-handed pump stroke shall displace the volume of air as specified.

The bag shall fully recover to its relaxed size and shape in not more than 1 second upon being released from the above stroke.

While being operated at a rate to produce an airflow of 15 LPM, a running test of valve ports pressure/negative pressure shall determine that the non-rebreathing valve does not allow rebreathing of exhausted air directly from the bag, and that all valves are operating smoothly and as intended.

The unit shall be operated in conjunction with an oxygen source. The low and high supplementary oxygen concentration shall be as specified.

The unit shall be fully disassembled. Attempts shall be made to assemble the unit incorrectly. No part shall be capable of assembly in a wrong location or incorrectly at its proper location.

The unit shall be measured and weighed to verify conformance with specifications.

Aging. Durability test shall be performed as specified.

Unit. Each (EA). One rescue breathing set, as specified, constitutes one unit

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U. S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

6515-00-890-1818

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY.

DoD - MB

Agent.

DLA - DM

CIVIL AGENCY COORDINATING ACTIVITY:

VA - OSS

Project No 6515-3547

Location H1-18-43

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